

Friday Session date 9/11/2026.

(1)

Geometric distribution: models # of trials until one success.

$$X \sim \text{Geom}(p) \quad P(X=k) = p(1-p)^{k-1}.$$

$$\begin{aligned} E[X] &= \sum_{k=1}^{\infty} k \cdot p(1-p)^{k-1} = p \sum_{k=1}^{\infty} k(1-p)^{k-1} = p \sum_{k=1}^{\infty} -\frac{\partial}{\partial p} (1-p)^k \\ &= p \frac{\partial}{\partial p} \sum_{k=1}^{\infty} (1-p)^k = p \left[-\frac{\partial}{\partial p} \left(\frac{1-p}{p} \right) \right] = p \left[-(-\frac{1}{p^2}) \right] = \frac{1}{p}. \end{aligned}$$

$$S = \frac{a_1}{1-r} \text{ for } a_n = r^n: \quad \sum_{k=1}^{\infty} (1-p)^k = \frac{1-p}{1-(1-p)} = \frac{1-p}{p}$$

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$$

Exercise: $\rightarrow$ why? how?

$$E[X(X-1)] + E[X] - (E[X])^2$$

let $q = 1-p$

$$= E[X(X-1)] + \frac{1}{p} - \left(\frac{1}{p}\right)^2$$

$$E[X(X-1)] = \sum_{k=1}^{\infty} k(k-1) p q^{k-1}$$

$$= p \sum_{k=1}^{\infty} k(k-1) q^{k-1} = p \sum_{k=1}^{\infty} \frac{\partial}{\partial q} (k-1) q^k = p \frac{\partial}{\partial q} \sum_{k=1}^{\infty} (k-1) q^k = p \frac{\partial}{\partial q} q^2 \sum_{k=1}^{\infty} (k-1) q^{k-2}$$

$$= p \frac{\partial}{\partial q} q^2 \sum_{k=2}^{\infty} \frac{\partial}{\partial q} q^{k-1} = p \frac{\partial}{\partial q} q^2 \frac{\partial}{\partial q} \sum_{k=2}^{\infty} q^{k-1} = p \frac{\partial}{\partial q} q^2 \frac{\partial}{\partial q} \sum_{k=1}^{\infty} q^k = p \frac{\partial}{\partial q} q^2 \frac{\partial}{\partial q} \frac{q}{1-q}$$

$$= p \frac{\partial}{\partial q} q^2 \frac{1}{(1-q)^2} = p \frac{\partial}{\partial q} \frac{q^2}{(1-q)^2} = p \frac{2q}{(1-q)^3} = \frac{2qp}{(1-q)^3}$$

$$\text{Var}[X] = \frac{2qp}{(1-q)^3} + \frac{1}{p} - \left(\frac{1}{p}\right)^2 = \frac{2(1-p)p}{(p)^3} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}$$

Poisson distribution.

$$X \sim \text{Poisson}(\lambda) \quad P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad k \Rightarrow \# \text{ of events happens.}$$

$$E[X] = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!} = \sum_{k=1}^{\infty} k \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=1}^{\infty} k \frac{\lambda^k}{k!}$$

Exercise: why $k=0$ here?

let $z = k-1$: $k=1, z=0$.

$$\otimes = \lambda e^{-\lambda} \sum_{z=0}^{\infty} \frac{\lambda^z}{z!}$$

$$= \lambda e^{-\lambda} e^{\lambda} = \lambda \cdot 1 = \lambda.$$

$$= \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{k \lambda^{k-1}}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{(k-1)}}{(k-1)!} = (*)$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad (\text{power series expansion})$$

$$\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2.$$

$$= E[X(X-1)] + E[X] - (E[X])^2.$$

$$= E[X(X-1)] + \lambda - \lambda^2.$$

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1) \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k}{k!}$$

$$= e^{-\lambda} \sum_{k=2}^{\infty} k(k-1) \frac{\lambda^k}{k!} = e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!} = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{(k-2)}}{(k-2)!}$$

let $z = k-2$.
 $k=2, z=0$.

$$= \lambda^2 e^{-\lambda} \sum_{z=0}^{\infty} \frac{\lambda^z}{z!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2.$$

$$\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

Gaussian:

$$X \sim N(\mu, \sigma^2) \quad f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

Moment:

$$\mathbb{E}[X^k] \leftarrow k\text{-th moment.}$$

Expectation is $\mathbb{E}[X] \Rightarrow 1\text{st moment.}$

Variance is $\mathbb{E}[(X-\mu)^2] \Rightarrow 2\text{nd } \underline{\text{Central}} \text{ moment.}$

For random variables with pdfs: $\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f(x) dx.$

Fun fact about Gaussian: If $X \sim N(0, \sigma^2)$:

$$\mathbb{E}[X^k] = 0 \text{ if } k \text{ is odd number.}$$

Exercise: why?

$\Uparrow$
(Go through it if there is time)

German Tank Problem;

Captured tanks from Nazi Germany, each tank is numbered with a serial number

Assume captured tanks has number 19, 40, 42, 60, 61, what is the total number of tanks produced?

Total: N tanks.

Question: Estimate N .

Captured: n tanks, $n \leq N$, each tank is named as X_i .

Let M be our estimator, m is a specific number, $M = \max\{X_1, X_2, \dots, X_n\}$.

$IP(M=m) = ? \Leftrightarrow$ Probability of us guessing $M=m$?

for $M=m$ to happen, we need: ① m -th tank is captured.

② the rest $n-1$ captured tanks must have serial number from $[1, m-1]$.

From ②, we have $\binom{m-1}{n-1}$ ways of capturing tanks that satisfies condition ②.

There are $\binom{N}{n}$ total ways of capturing tanks.

$$\text{So } IP(M=m) = \frac{\binom{m-1}{n-1}}{\binom{N}{n}}.$$

$$E[M] = \sum_{m=n}^N m \cdot \frac{\binom{m-1}{n-1}}{\binom{N}{n}} = \sum_{m=n}^N \frac{m \cdot (m-1)!}{(n-1)! (m-n)!} / \binom{N}{n}.$$

Exercise: why n ?

$$= \frac{1}{\binom{N}{n}} \sum_{m=n}^N \frac{m!}{(m-n)! (n-1)!} = \frac{1}{\binom{N}{n}} \sum_{m=n}^N \frac{m!}{(m-n)! n!} = \frac{1}{\binom{N}{n}} \sum_{m=n}^N n \cdot \binom{m}{n}.$$

$$= \frac{n}{\binom{N}{n}} \sum_{m=n}^N \binom{m}{n} = \frac{n}{\binom{N}{n}} \binom{N+1}{n+1} = \frac{n(N+1)}{n+1}$$

Exercise: How?

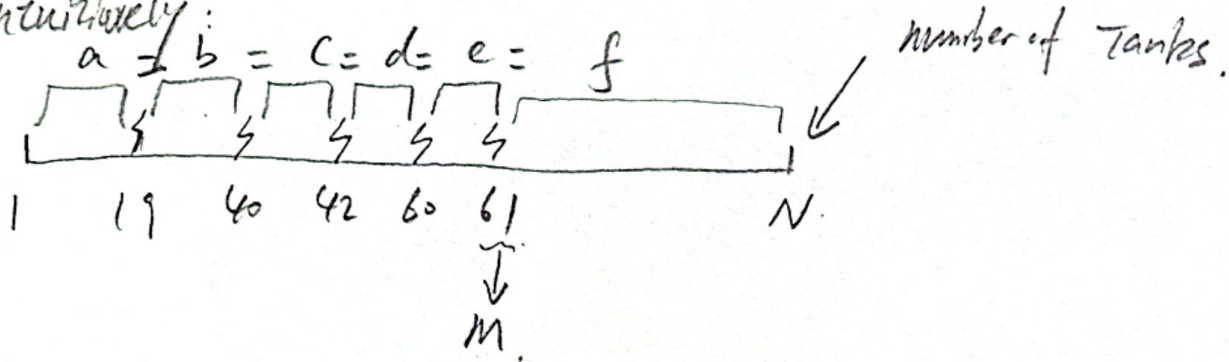
$$\mathbb{E}[M] = \frac{n(N+1)}{n+1} \Rightarrow N - \mathbb{E}[M] = \frac{N-n}{n+1} \Leftarrow \begin{matrix} \text{How much} \\ \text{are we missing} \\ \text{on average} \end{matrix} \quad (3)$$

$\Downarrow$

$$N = \frac{(n+1)\mathbb{E}[M] - n}{1}, \text{ replace } \mathbb{E}[M] \text{ with } M:$$

$$\hat{N} = \frac{(n+1)M - n}{1} = \frac{n+1}{1} M - 1$$

Intuitively:



"Assumed" the gaps are equal length.

Based on what we know: the "gap" is about $\frac{M}{n} = \frac{61}{5} \approx 12$.

$$(n+1) \cdot 12 = \# \text{ of total "gaps"} \cdot \text{"gap length"} = 6 \cdot 12 = \underline{72}$$

$$\hat{N} = \frac{(n+1)M - n}{1} = \underline{72.2} \quad \underline{\underline{\text{Not that far!}}}$$